SERAPHIM LQG FRAMEWORK

Grand Total Claims, Predictions & Evidence Audit

Based on: v18.4 Main Paper · Paper 2 v16 · Compact Binary Degeneracy v3

lis10inc · Independent Researcher · April 2026 · Personal Working Reference

GRAND TOTAL: 63 items across 5 confidence tiers + 5 open questions

C1–C19 19 Confirmed D1–D12 12 Derived P1–P14 14 Pending S1–S9 9 Suggestive Sp1–Sp4 4 Speculative OQ1–OQ5 5 Open Questions

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Core equation: $n(C) = 3.561 + 3.506 \times C$

$K_o = 1.1467 \times 10^{84}$ Hz² [CRITICAL: exponent is 84, not 4. Unicode corruption risk in rendered text.]

Methodology and Honest Accounting

This document is a complete audit of every claim, prediction, and open question across the Seraphim LQG Framework as of v18.4 (April 2026). Every item is assigned exactly one confidence tier. No result is promoted to a higher tier than the evidence supports. No result is buried.

The audit draws from three source papers:

v18.4 — Main paper (264 BBH events, nine independent tests, four integrated companion derivations)

Paper 2 v16 — Octave Hierarchy (speculative extensions, periodic table, phase grids)

Compact Binary Degeneracy v3 — Population spectrum (BNS, NSBH, EMRI, Casimir connection)

TIER 1 — CONFIRMED

Empirically tested and passed. These results have been tested against real data. They passed. They can still be falsified by new data, but the current evidence is affirmative.

ID	Claim	Evidence	Precision
C1	n_BBH = 5.314 from zero free parameters	249 BBH, 3 catalogs	0.05σ (Δ=+0.005)
C2	BBH band stable across all 3 catalogs independently	Means: 5.329 / 5.298 / 5.329	All within ±0.016
C3	Robertson confirmed over Heisenberg	Pop. mean 5.319 vs 4.314	1.000 oct separation
C4	Waveform independence (IMRPhenomXPHM vs	Per-event GWTC-2.1 comparison	Δ = 0.070%

ID	Claim	Evidence	Precision
	SEOBNRv4PHM)		
C5	Redshift invariance dn/dz = 0 in BBH band	Spearman r=0.010, p=0.875, N=248	z = 0.04–2.15
C6	q and χ_eff carry independent signals (partial corr.)	partial r > 0.60, all catalogs	$p < 10^{-8}$
C7	n_flip = 3.561 is a hard floor — zero violations	264 events, 3 catalogs	0 violations
C8	Coupling ladder terminates at 1/α = 137.036	Intercepts 132.97→135.00→137.036	Step = 2.03
C9	Gap mass-scaling: gap = 126.318 + $\log_2(M/M_\odot)$	264 events; slope 0.922 explained	$r = 0.978$, $p < 10^{-58}$
C10	Tiling identity $N_\odot \cdot A_j = \pi(\Delta x)^2$ is algebraic identity	Symbolic: γ and ℓ_P² cancel exactly	12 sig. figures
C11	H condensation ceiling n_H = 92.19 sorts all 118 elements	Zero violations, periodic table	0 free params
C12	GW190425 tidal deformability range check passes	n_pred = 4.11–4.19, in range	Partial check
C13	β = 0.814 compactness exponent stable	6 × 1M Monte Carlo runs	95% CI [0.75, 0.88]
C14	Helix winding W = 1/α = 137.036	N_Hub / p derivation	Exact to 8 d.p.
C15	H_0_local = 73.96 km/s/Mpc from Immirzi mismatch	SH0ES+JWST 2025: 73.8 ± 0.88	0.18σ residual
C16	χ_p (precessing spin) null at q > 0.70	r ≈ 0.000, p = 0.99 near-equal-mass	Confirmed null
C17	b_eta = 0.698 after spin control, ~0.5% from d_eff=5/2	Test 18, N=167 in-band BBH	$R^2 = 0.948$
C18	χ_eff × $\log_2\eta$ interaction confirmed	$p = 10^{-37}$, ΔAIC = −163 points	Test 20
C19	b_eta decreases monotonically with mass (20→100+ $M_\odot$)	Tests 21; range 0.90 to 0.55	Quantum→classical

NOTE: C15 (Hubble tension): k_seesaw = 11 is derived by two independent paths (desert wall arithmetic and Robertson cascade) but the derivation is not airtight. The 0.18σ match is the strongest current number. Euclid 2026 will harden or break this.

NOTE: C10 (Tiling identity): This is a mathematical proof, not a measurement. It removes one previously listed 'postulate' from the framework. Confirmed in the sense that it has been verified symbolically and numerically.

TIER 2 — DERIVED

Mathematically forced from confirmed constants; no free parameters. These results follow algebraically or analytically from the confirmed framework constants. They make no new assumptions. Falsifiable when O5 data arrives or LISA flies.

ID	Claim	Derivation path	Status
D1	j = 1/2 from spin foam minimum action on Robertson surface	S_total monotone in j; min at j=1/2. HONEST: free EPRL peaks at j*=0.951; Robertson constraint required.	Derived
D2	n_flip = 3.561 from γ_entropy + hadronic correction	n_flip = 0.9099 + (n_QCD − n_proton) = 3.560; Δ = 0.001 oct (0.04%)	Derived
D3	K_0 = 1.1467 × 10^84 Hz² from fundamental constants	c²/(128π²·γ_area·ℓ_P²); verified numerically	Derived
D4	C = +0.5 from Robertson minimum alone (no GR input)	Robertson Δx·Δp = ħ/2 → M = m_P/2 → C = 1/2 exactly	Derived
D5	C = −0.5 anti-horizon (de Sitter) from Robertson	Geometric realm direction; same algebra as D4	Derived
D6	Phase grid offset = 0.055 = 3n_flip − 2n_BBH	Algebraic identity from confirmed anchors	Exact
D7	IMBH gap ≈ 1/α at M_0 = 1684 M☉	gap = 1/α when M = 2^(137.036 − 126.318) M☉	Pending LISA/ET
D8	EMRI limit: n → n_flip as q → 0 (monotone)	n_EMRI = 3.561 + 7.012η → 3.561	LISA prediction
D9	BNS n = C_NS directly for equal-mass (4η = 1 cancels)	4 × 0.25 × C_NS = C_NS exactly	O5 pending
D10	Degeneracy condition: η_BBH = C_NS/2 → identical n	Algebraic; GW190814: Δ = 0.003 oct	Template confirmed
D11	Slope 3.506 = χ(S²) × Δn (topology forced)	2 × 1.753 = 3.506; Euler characteristic of S²	Structural identity
D12	k_seesaw = 11 by two independent paths	Path 1: (3+8)/2 × 2 = 11. Path 2: Robertson cascade ~11.125. Both	Converging

ID	Claim	Derivation path	Status
		prime.	

TIER 3 — PENDING FALSIFICATION

Predicted, testable, not yet tested against data. Each one has a clear observable, a falsification condition, and a timeline. O5 is the primary test run.

ID	Prediction	Falsified if…	Timeline
P1	O5 BBH events cluster at n = 5.314 ± 0.15	Population mean outside ±0.15	O5
P2	O5 BNS: slope of n vs C_NS = 3.506 confirmed	Slope absent or ≠ 3.506	O5
P3	BNS band n = 4.05–4.27 (all EOS, equal mass)	BNS cluster outside predicted band	O5
P4	NSBH at n = 3.561 + 2.525η (below BNS band)	NSBH cluster at n ≥ 4.05	O5
P5	No O5 event at n < n_flip = 3.561	Single confirmed event below floor	O5 (hardest test)
P6	β → 1.0 in O5 limit (current: β = 0.814)	β stable at 0.814 with high-precision O5 tidal data	O5
P7	Discriminate d_eff=5/2 (b=0.7012) vs BEC (b=0.750)	CI width requires ~1505 events; O5 narrows by ~40%	O5+O6
P8	b_eta → 0.5 at LISA masses M ~ 10^5–10^7 M☉	b_eta constant across all mass scales	LISA
P9	IMBH mergers at M ≈ 1684 M☉ show gap ≈ 1/α	Gap ≠ 137 at IMBH mass scale	LISA/ET/CE
P10	EMRIs approach n_flip monotonically, never cross it	EMRI population at n < 3.561	LISA
P11	H_0_local = 73.96 km/s/Mpc to ±0.3 km/s/Mpc	Euclid/SKAO outside 73.96 ± 0.3	Euclid 2026
P12	ADMX axion at chi k=56, E = 1.25 μeV	ADMX scan clears this frequency with no signal	Near-term
P13	Realm-projection geometry universal for compact objects	O5 NSBH/BNS C mismatch with n(C)	O5

ID	Prediction	Falsified if…	Timeline
P14	PTA spectral index constrains LQG at n = 173 scale	NANOGrav/PPTA inconsistent with LQG patch geometry	Ongoing

NOTE: P7: PN leading order b = 0.5 is already excluded at 7.5σ. The live contest is d_eff=5/2 (b=0.7012) vs BEC universality (b=0.750). Both inside current CI. O6 decides.

NOTE: P5: The floor has not been violated in 264 events. Any single event below n_flip = 3.561 in O5 requires either a physical explanation (new quantum geometry regime) or indicates a flaw in the tiling postulate.

TIER 4 — SUGGESTIVE

Real signal present; mechanism not proven. These findings show numerically interesting patterns — too precise to be random noise, but without a proved mechanism linking them to the core framework. Documented. Not promoted.

ID	Finding	Evidence	What's missing
S1	Coupling ladder step ≈ $\chi(S^2)$ = 2.03	3 catalogs, r=0.892, $p=2.8\times10^{-87}$	NR waveform systematics not ruled out
S2	n_BBH / n_Hubble = 1/38 to 0.03%	Single rational harmonic, 3 anchors	No mechanism for the 38
S3	n_LIGO = (19/28)·n_Hubble to 0.08%	Single rational harmonic	No mechanism
S4	Seesaw scale = Grid 2 rung k=11, Δ = 0.008 oct	Sub-grid-step precision on one scale	k=11 mechanism open
S5	EW/BBN transitions: 3.5× excess of ladder-step hits	6 of 21 pairs (expected 1.7)	Pattern real; physics unclear
S6	Casimir window (89–96) = metallization cluster	Scale coincidence with known physics	No dynamical derivation
S7	LQG horizon and Casimir plate: same d_eff = 5/2 surface	3 independent measurements converge	Same mechanism not proved
S8	GUT desert = interstitial space between Grid 1 and Grid 2	String k=3 (Δ=0.019), seesaw k=8 (Δ=0.011)	Desert emptiness not analytically derived
S9	Matter-forming epoch clusters near 202-oct cavity center	16-oct window near midpoint; Δn=3.0 from exact center	Δn=3 is not negligible

TIER 5 — SPECULATIVE

Labeled speculative in the source papers. May be directionally correct or coincidental. Documented because they emerged from the framework's internal logic. Not used as evidence for anything else.

ID	Claim	Basis	Why speculative
Sp 1	Mirror sector ($C < 0$ reflection) = dark matter candidate	GUP argument: N★ < 1 in mirror; no photons, no gravitons	No observational path yet; GUP argument not fully formalised
Sp 2	Cosmological constant as 202-octave cavity resonance	Anti-horizon $C=-0.5$; 60-order gap = 114 harmonics of Seraphim interval	No mechanism; harmonic coincidence unproved
Sp 3	Metallic hydrogen under pressure tests $d_{eff} = 5/2 \to 2$ collapse	Physically realizable; diamond anvil cell + Casimir measurement	Not yet designed or run as an experiment
Sp 4	Riemann zeros map to spectral boundaries in octave hierarchy	Sidequest observation; not developed into a testable claim	No framework connection established

OPEN QUESTIONS

Explicitly unresolved as of v18.4. No claim is made about them. Recorded because honesty requires knowing where the edges are.

ID	Question	Where it matters	What would close it
OQ 1	Formal LQG spin foam derivation of prime-indexing rule $k = p(\dim(j))$	Connects ladder rungs to prime numbers	Full EPRL derivation on Robertson surface
OQ 2	Exact baryon-loading path in Hubble tension master identity	$k_{seesaw} = 11$ derivation not fully closed	Analytical derivation of hadronic correction path
OQ 3	Why $n_{Hubble} \approx 202$? (Hubble scale not derived from LQG + Robertson)	Paper 2 hierarchy; N_{Hub} used but not derived	First-principles derivation of Hubble horizon in octave units
OQ 4	Whether rational harmonics 1/38 and 19/28 have topological origin	S2 and S3 suggestive results	Derivation of mode numbers from Euler characteristic or spin network

ID	Question	Where it matters	What would close it
OQ 5	Why the GUT desert (n=9.58–19.55) is structurally empty	S8; Grid 1/Grid 2 gap description	Analytical proof that neither phase grid has a rung in this range

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Grand Total Scorecard

Every item across all three papers, counted once, assigned to exactly one tier.

Tier	Count	Items	Primary test event
CONFIRMED	19	C1–C19	Already tested (GWTC-2.1/3/4, SH0ES+JWST 2025)
DERIVED	12	D1–D12	Structural; falsifiable by O5 population
PENDING FALSIFICATION	14	P1–P14	O5 (primary), LISA (secondary), Euclid 2026
SUGGESTIVE	9	S1–S9	O5 / LISA / deeper theoretical work
SPECULATIVE	4	Sp1–Sp4	No firm timeline; experimental or theoretical
OPEN QUESTIONS	5	OQ1–OQ5	Theoretical derivation work
TOTAL	63	—	—

Key Risk Summary

The three results most likely to face reviewer scrutiny, in order of vulnerability:

1. β = 0.814 (C13) — Empirically stable across 6 × 1M runs but β = 1.0 is predicted in the O5 limit. The gap between current data and theory is real and noted. The live contest between d_eff=5/2 (b=0.7012) and BEC (b=0.750) requires ~1505 events to decide. Current CI covers both.

2. k_seesaw = 11 in Hubble tension (C15/D12) — Two independent derivation paths converge, but the convergence is not fully tight (Path 2 gives 11.125, not 11.000). The 0.18σ H_0 match is striking. Euclid 2026 will decide.

3. j = 1/2 spin foam derivation (D1) — HONEST NOTE: the free EPRL amplitude peaks at j* = 0.951, not j = 1/2. The Robertson constraint surface is required to select j = 1/2. This is a constrained, not a free, minimum-action result.

Primary Falsification Event

O5 LIGO run is the decisive test. Predictions that will either confirm or falsify the framework in one run:

P1 — BBH band at $n = 5.314 \pm 0.15$ (extends C1 from 264 to ~750+ events)

P3 — BNS cluster at $n = 4.05$–4.27 (first new binary class test)

P4 — NSBH below BNS band (tests NSBH formula)

P5 — No event below n_flip = 3.561 (extends floor test to ~500+ new events)

P7 — b_eta narrows toward 0.7012 or 0.750 (O5 narrows CI ~40%)
